

This is a remarkably elegant concept, and **it is a mathematically perfect match!**

When Charles Janet created his Left-Step Periodic Table in 1928, he designed it to end at **Element 120** (Unbinilium, $Z=120$). Even though only 92 elements were known at the time, Janet realized that quantum orbital mechanics naturally creates a closed, 120-element shell system. Pairing this table with a **disdyakis triacontahedron** (a 120-sided polyhedron, or **d120**) links quantum chemistry directly to 3D geometry.

The Quantum Math Behind 120

In Janet's Left-Step table, elements are ordered strictly by the filling of electron subshells following the Madelung rule ($n + l$). The periods come in pairs based on orbital blocks (s, p, d, f):

- **Period 1 & 2 (s-block only):** $2 + 2 = 4$ elements (1s, 2s)
- **Period 3 & 4 (s, p blocks):** $8 + 8 = 16$ elements (2p, 3s, 3p, 4s)
- **Period 5 & 6 (s, p, d blocks):** $18 + 18 = 36$ elements (3d, 4p, 5s, 4d, 5p, 6s)
- **Period 7 & 8 (s, p, d, f blocks):** $32 + 32 = 64$ elements (4f, 5d, 6p, 7s, 5f, 6d, 7p, 8s)

$\text{Total Elements} = 2(1^2 + 1^2 + 2^2 + 2^2 + 3^2 + 3^2 + 4^2 + 4^2) = 120$

Element 120 ($8s^2$) represents the exact natural boundary where the f-block period sequence completes.

How to Map the 120 Elements onto the d120 Polyhedron

A disdyakis triacontahedron has full **icosahedral symmetry (I_h)**, which consists of exactly **120 symmetry operations**. This means every single face is identical in shape and can be systematically mapped to one element:

[North Pole / Top Axis]

- s-block (16 faces around 12-fold vertices)
- p-block (36 faces along 6-fold vertices)
- d-block (40 faces along 4-fold edges)
- f-block (28 faces around Equatorial belts)

[South Pole / Bottom Axis]

Mapping Strategy by Orbital Block

1. **Latitude as Period ($n+l$ sum):**
 - **Element 1 (Hydrogen)** at the North Pole.
 - **Element 120 (Unbinilium)** at the South Pole.
 - Elements spiral down in atomic number $Z=1$ to 120 around the polyhedron, creating a **3D quantum globe**.
2. **Face Color-Coding by Subshell (s, p, d, f):**
 - **16 s-block faces** (Alkali/Alkaline Earth metals) grouped near the poles.
 - **36 p-block faces** (Non-metals, Halogens, Noble gases).
 - **40 d-block faces** (Transition metals) forming mid-latitude rings.
 - **28 f-block faces** (Lanthanides and Actinides) spanning the equatorial region.

Practical Applications for a 120-Element Solid

- 1. The Ultimate Chemistry Die (The Element d120)** Roll the die to pick random elements for chemistry challenges, game mechanics, or teaching atomic properties.
- 2. Interactive Quantum Globe** An antipodal structure where opposite faces could highlight nuclear stability, decay chains, or electron valence pairings.
- 3. Tactile 3D Periodic Sculpture** A physical model where magnetic face tiles allow students to assemble the universe's building blocks in orbital order.

To map all 120 elements of the Left-Step Janet Periodic Table onto a disdyakis triacontahedron (d120), we exploit the polyhedron's full icosahedral symmetry (I_h).

A d120 is made of 120 identical right-angled scalene triangular faces. Because its symmetry group contains exactly 120 symmetry operations (reflections and rotations), every single face maps 1-to-1 with an element, creating a mathematically unified 3D Quantum Globe.

Polyhedral Geometry & Vertex Hierarchy

The 120 faces are arranged around 62 total vertices divided into three distinct types:

- * 12 Vertices with order-10 degree (Vertices where 10 triangular faces meet) $\rightarrow$ Top/Bottom Pole Regions

- * 20 Vertices with order-6 degree (Vertices where 6 faces meet) $\rightarrow$ Mid-Latitude Regions

- * 30 Vertices with order-4 degree (Vertices where 4 faces meet) $\rightarrow$ Equatorial Belt

The Mapping Scheme: Spiral Quantum Latitudes

The 120 faces naturally partition into 8 horizontal latitude rings (Periods 1 to 8 of Janet's Table). Within each ring, faces are ordered sequentially by atomic number Z, spiraling clockwise from top to bottom.

[NORTH POLE: Z=1 (H)]

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Ring 1 & 2: s-block (Z=1 to 4) $\rightarrow$ Order-10 Vertex

Ring 3 & 4: s, p blocks (Z=5 to 20) $\rightarrow$ Order-6 Vertices

Ring 5 & 6: s, p, d blocks (Z=21 to 56) $\rightarrow$ Order-4 / Order-6 Vertices

Ring 7 & 8: f, d, p, s blocks (Z=57 to 120) $\rightarrow$ Equatorial Belt

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[SOUTH POLE: Z=120 (Ubn)]

Face Allocation by Period & Subshell

| Period (Janet Table) | Orbital Subshells Filled | Face Count | Atomic Numbers (Z) | Geometric Region on d120 |

|---|---|---|---|---|

| Period 1 | 1s | 2 | Z = 1 - 2 | Top polar cap surrounding North Vertex |

| Period 2 | 2s | 2 | Z = 3 - 4 | Upper polar ring below 1s cap |

| Period 3 | 2p, 3s | 8 | Z = 5 - 12 | Northern high-latitude ring |

| Period 4 | 3p, 4s | 8 | Z = 13 - 20 | Northern mid-latitude ring |

| Period 5 | 3d, 4p, 5s | 18 | Z = 21 - 38 | Northern temperate belt |

| Period 6 | 4d, 5p, 6s | 18 | Z = 39 - 56 | Northern equatorial transition |

| Period 7 | 4f, 5d, 6p, 7s | 32 | Z = 57 - 88 | Equatorial belt (Upper half) |

| Period 8 | 5f, 6d, 7p, 8s | 32 | Z = 89 - 120 | Southern equatorial to South Pole |

Orbital Block Color-Coding Scheme

To make the die intuitive to read, each of the four quantum blocks (s, p, d, f) gets a distinct base color:

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	[s-block]	RED		16 faces Alkali & Alkaline Earths	
	[p-block]	YELLOW		36 faces Post-transition, Halogens	
	[d-block]	BLUE		40 faces Transition Metals	
	[f-block]	GREEN		28 faces Lanthanides & Actinides	
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* Red (s-block): 16 faces

* Placed along the polar axis (Periods 1–2) and trailing edge of every latitude ring (8s ends at Z=120 at the exact South Pole).

* Yellow (p-block): 36 faces

* Concentrated in 6-fold vertex clusters around the upper and lower temperate zones.

* Blue (d-block): 40 faces

* Arranged around the 4-fold vertex chains bridging the mid-latitudes.

* Green (f-block): 28 faces

* Form two dense 14-element rings straddling the equator.

Symmetry Properties & Game Features

> Antipodal Pairing (Z vs. 121 - Z):

> Because a d120 has central symmetry (every face has an exact opposite face), face Z is directly opposite face 121 - Z.

> * Z = 1 (Hydrogen) is directly opposite Z = 120 (Unbinilium).

> * Z = 6 (Carbon) is directly opposite Z = 115 (Moscovium).

> * Z = 26 (Iron) is directly opposite Z = 95 (Americium).

> This guarantees that rolling the die pairs the lightest/simplest quantum states against the heaviest theoretical states!

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Here is the complete face-mapping table for all 120 elements onto the disdyakis triacontahedron (d120).

In this mapping, the Face Index matches atomic number Z, spiraling sequentially down the polyhedron from the North Pole (Z=1) to the South Pole (Z=120). Periods are assigned according to the Janet Left-Step Periodic Table (n+l filling order), and each face's Antipodal Element corresponds to the geometrically opposite face on the die (121 - Z).

The Complete d120 Quantum Mapping Table

| Face Index (Z) | Element Name | Symbol | Janet Period | Subshell | Orbital Block | Antipodal Face (121-Z) | Antipodal Element |

|---|---|---|---|---|---|---|

1	Hydrogen	H	1	1s	s	120	Unbinilium (Ubn)
2	Helium	He	1	1s	s	119	Ununennium (Uue)
3	Lithium	Li	2	2s	s	118	Oganesson (Og)
4	Beryllium	Be	2	2s	s	117	Tennessine (Ts)
5	Boron	B	3	2p	p	116	Livermorium (Lv)
6	Carbon	C	3	2p	p	115	Moscovium (Mc)
7	Nitrogen	N	3	2p	p	114	Flerovium (Fl)
8	Oxygen	O	3	2p	p	113	Nihonium (Nh)

9	Fluorine	F	3	2p	p	112	Copernicium (Cn)
10	Neon	Ne	3	2p	p	111	Roentgenium (Rg)
11	Sodium	Na	3	3s	s	110	Darmstadtium (Ds)
12	Magnesium	Mg	3	3s	s	109	Meitnerium (Mt)
13	Aluminium	Al	4	3p	p	108	Hassium (Hs)
14	Silicon	Si	4	3p	p	107	Bohrium (Bh)
15	Phosphorus	P	4	3p	p	106	Seaborgium (Sg)
16	Sulfur	S	4	3p	p	105	Dubnium (Db)
17	Chlorine	Cl	4	3p	p	104	Rutherfordium (Rf)
18	Argon	Ar	4	3p	p	103	Lawrencium (Lr)
19	Potassium	K	4	4s	s	102	Nobelium (No)
20	Calcium	Ca	4	4s	s	101	Mendelevium (Md)
21	Scandium	Sc	5	3d	d	100	Fermium (Fm)
22	Titanium	Ti	5	3d	d	99	Einsteinium (Es)
23	Vanadium	V	5	3d	d	98	Californium (Cf)
24	Chromium	Cr	5	3d	d	97	Berkelium (Bk)
25	Manganese	Mn	5	3d	d	96	Curium (Cm)
26	Iron	Fe	5	3d	d	95	Americium (Am)
27	Cobalt	Co	5	3d	d	94	Plutonium (Pu)
28	Nickel	Ni	5	3d	d	93	Neptunium (Np)
29	Copper	Cu	5	3d	d	92	Uranium (U)
30	Zinc	Zn	5	3d	d	91	Protactinium (Pa)
31	Gallium	Ga	5	4p	p	90	Thorium (Th)
32	Germanium	Ge	5	4p	p	89	Actinium (Ac)
33	Arsenic	As	5	4p	p	88	Radium (Ra)
34	Selenium	Se	5	4p	p	87	Francium (Fr)
35	Bromine	Br	5	4p	p	86	Radon (Rn)
36	Krypton	Kr	5	4p	p	85	Astatine (At)
37	Rubidium	Rb	5	5s	s	84	Polonium (Po)
38	Strontium	Sr	5	5s	s	83	Bismuth (Bi)
39	Yttrium	Y	6	4d	d	82	Lead (Pb)
40	Zirconium	Zr	6	4d	d	81	Thallium (Tl)
41	Niobium	Nb	6	4d	d	80	Mercury (Hg)
42	Molybdenum	Mo	6	4d	d	79	Gold (Au)
43	Technetium	Tc	6	4d	d	78	Platinum (Pt)
44	Ruthenium	Ru	6	4d	d	77	Iridium (Ir)
45	Rhodium	Rh	6	4d	d	76	Osmium (Os)
46	Palladium	Pd	6	4d	d	75	Rhenium (Re)
47	Silver	Ag	6	4d	d	74	Tungsten (W)
48	Cadmium	Cd	6	4d	d	73	Tantalum (Ta)
49	Indium	In	6	5p	p	72	Hafnium (Hf)
50	Tin	Sn	6	5p	p	71	Lutetium (Lu)
51	Antimony	Sb	6	5p	p	70	Ytterbium (Yb)
52	Tellurium	Te	6	5p	p	69	Thulium (Tm)
53	Iodine	I	6	5p	p	68	Erbium (Er)
54	Xenon	Xe	6	5p	p	67	Holmium (Ho)
55	Caesium	Cs	6	6s	s	66	Dysprosium (Dy)
56	Barium	Ba	6	6s	s	65	Terbium (Tb)

57	Lanthanum	La	7	4f	f	64	Gadolinium (Gd)
58	Cerium	Ce	7	4f	f	63	Europium (Eu)
59	Praseodymium	Pr	7	4f	f	62	Samarium (Sm)
60	Neodymium	Nd	7	4f	f	61	Promethium (Pm)
61	Promethium	Pm	7	4f	f	60	Neodymium (Nd)
62	Samarium	Sm	7	4f	f	59	Praseodymium (Pr)
63	Europium	Eu	7	4f	f	58	Cerium (Ce)
64	Gadolinium	Gd	7	4f	f	57	Lanthanum (La)
65	Terbium	Tb	7	4f	f	56	Barium (Ba)
66	Dysprosium	Dy	7	4f	f	55	Caesium (Cs)
67	Holmium	Ho	7	4f	f	54	Xenon (Xe)
68	Erbium	Er	7	4f	f	53	Iodine (I)
69	Thulium	Tm	7	4f	f	52	Tellurium (Te)
70	Ytterbium	Yb	7	4f	f	51	Antimony (Sb)
71	Lutetium	Lu	7	5d	d	50	Tin (Sn)
72	Hafnium	Hf	7	5d	d	49	Indium (In)
73	Tantalum	Ta	7	5d	d	48	Cadmium (Cd)
74	Tungsten	W	7	5d	d	47	Silver (Ag)
75	Rhenium	Re	7	5d	d	46	Palladium (Pd)
76	Osmium	Os	7	5d	d	45	Rhodium (Rh)
77	Iridium	Ir	7	5d	d	44	Ruthenium (Ru)
78	Platinum	Pt	7	5d	d	43	Technetium (Tc)
79	Gold	Au	7	5d	d	42	Molybdenum (Mo)
80	Mercury	Hg	7	5d	d	41	Niobium (Nb)
81	Thallium	Tl	7	6p	p	40	Zirconium (Zr)
82	Lead	Pb	7	6p	p	39	Yttrium (Y)
83	Bismuth	Bi	7	6p	p	38	Strontium (Sr)
84	Polonium	Po	7	6p	p	37	Rubidium (Rb)
85	Astatine	At	7	6p	p	36	Krypton (Kr)
86	Radon	Rn	7	6p	p	35	Bromine (Br)
87	Francium	Fr	7	7s	s	34	Selenium (Se)
88	Radium	Ra	7	7s	s	33	Arsenic (As)
89	Actinium	Ac	8	5f	f	32	Germanium (Ge)
90	Thorium	Th	8	5f	f	31	Gallium (Ga)
91	Protactinium	Pa	8	5f	f	30	Zinc (Zn)
92	Uranium	U	8	5f	f	29	Copper (Cu)
93	Neptunium	Np	8	5f	f	28	Nickel (Ni)
94	Plutonium	Pu	8	5f	f	27	Cobalt (Co)
95	Americium	Am	8	5f	f	26	Iron (Fe)
96	Curium	Cm	8	5f	f	25	Manganese (Mn)
97	Berkelium	Bk	8	5f	f	24	Chromium (Cr)
98	Californium	Cf	8	5f	f	23	Vanadium (V)
99	Einsteinium	Es	8	5f	f	22	Titanium (Ti)
100	Fermium	Fm	8	5f	f	21	Scandium (Sc)
101	Mendelevium	Md	8	5f	f	20	Calcium (Ca)
102	Nobelium	No	8	5f	f	19	Potassium (K)
103	Lawrencium	Lr	8	6d	d	18	Argon (Ar)
104	Rutherfordium	Rf	8	6d	d	17	Chlorine (Cl)

105	Dubnium	Db	8	6d	d	16	Sulfur (S)
106	Seaborgium	Sg	8	6d	d	15	Phosphorus (P)
107	Bohrium	Bh	8	6d	d	14	Silicon (Si)
108	Hassium	Hs	8	6d	d	13	Aluminium (Al)
109	Meitnerium	Mt	8	6d	d	12	Magnesium (Mg)
110	Darmstadtium	Ds	8	6d	d	11	Sodium (Na)
111	Roentgenium	Rg	8	6d	d	10	Neon (Ne)
112	Copernicium	Cn	8	6d	d	9	Fluorine (F)
113	Nihonium	Nh	8	7p	p	8	Oxygen (O)
114	Flerovium	Fl	8	7p	p	7	Nitrogen (N)
115	Moscovium	Mc	8	7p	p	6	Carbon (C)
116	Livermorium	Lv	8	7p	p	5	Boron (B)
117	Tennessine	Ts	8	7p	p	4	Beryllium (Be)
118	Oganesson	Og	8	7p	p	3	Lithium (Li)
119	Ununennium	Uue	8	8s	s	2	Helium (He)
120	Unbinilium	Ubn	8	8s	s	1	Hydrogen (H)